

SCAM-Net: Unifying Box Annotations and Pixel-Level Masks for Fundus Image Segmentation

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Abstract. Accurate segmentation of the optic cup and optic disc is crucial for glaucoma diagnosis through the vertical cup-to-disc ratio. Prevailing fully-supervised deep learning methods, however, rely on pixel-level annotations, which are labor intensive and costly to obtain. While weakly supervised approaches leveraging foundational models like SAM with bounding box prompts reduce this dependency, they often yield sub-optimal performance on fundus images due to the inherently ambiguous boundaries of ocular structures. This work addresses the challenging task of generating high-quality segmentation masks from sparse bounding-box annotations. We propose SCAM-Net, a novel instance segmentation framework designed for a practical mixed-supervision setting, which synergistically utilizes both limited pixel-level ground truth and abundant, cost-effective box annotations for accurate optic cup and optic disc segmentation. Our framework introduces a high-performance pseudo-label generation strategy that fuses the semantic localization from Class Activation Maps with the fine-grained details from the Segment Anything Model, effectively translating sparse annotations into dense, high-fidelity pixel-level guidance. Furthermore, we introduce a pixel-level contrastive learning mechanism that leverages these generated pseudo masks as priors. By explicitly contrasting features between foreground and background regions, our method enhances feature discriminability, significantly improving segmentation accuracy at the challenging, low-contrast optical cup and optic disc boundaries. Extensive experiments on three public datasets, Drishti-GS, RIM-ONE, and REFUGE, demonstrate that SCAM-Net substantially outperforms existing state-of-the-art methods. By achieving precise segmentation with easily accessible bounding box annotations, our work presents a cost-effective and highly efficient solution for automated glaucoma screening.

Keywords: Segment Anything · Partially-Supervised · Pseudo Label · Contrastive Learning · Optic Disc Segmentation.

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1 INTRODUCTION

Glaucoma is a leading cause of irreversible blindness, progressing silently and posing a major public health challenge. Early detection depends on structural assessment of the Optic Nerve Head (ONH), with the Vertical Cup-to-Disc Ratio (vCDR) serving as a key biomarker [31]. Accurate segmentation of the optic disc (OD) and optic cup (OC) in fundus images is essential for automated vCDR calculation and objective glaucoma diagnosis. Although fully supervised deep learning methods have achieved impressive results [37], their dependence on pixel-level manual annotations is time-consuming, expensive, and restricts the development of large-scale, high-quality datasets.

To address annotation costs, weakly and semi-supervised learning have gained traction in medical image segmentation. Weakly supervised methods use coarse labels (e.g., image-level, bounding boxes [11]), but often produce pseudo masks that lack fine boundaries, especially in fundus images where OD/OC boundaries are ambiguous, leading to significant segmentation errors [21]. Semi-supervised methods leverage a small set of labeled data and abundant unlabeled data with strategies like pseudo-labeling [24] and contrastive learning, reducing annotation costs while maintaining accuracy. However, they typically assume labeled and unlabeled data are from the same domain, which is often not the case in clinical practice due to variations in imaging devices and patient populations, limiting generalization.

Recently, pseudo-label generation using pre-trained models has shown promise. CAM-based methods with ImageNet-pretrained backbones can generate pseudo masks, but their spatial resolution and boundary details are limited, making them insufficient for medical image segmentation. The Segment Anything Model (SAM) [27] has gained attention for its strong zero-shot generalization [18] and ability to generate high-quality masks. However, in fundus images, low contrast and close OD/OC boundaries make prompts noisy, degrading mask quality. Some studies [8] have refined SAM-generated pseudo-labels with contrastive learning, but these often require fine-tuning or domain adaptation, and SAM’s computational demands limit clinical scalability [32].

To address these challenges, we propose SCAM-Net, a novel instance segmentation network based on mixed supervision for precise OD/OC segmentation. SCAM-Net leverages both box annotations and pixel-level masks, and consists of a Dual-Pyramid Feature Extraction Module, a Mixed-Supervision-based Pseudo-Mask Generation Module, and a Unified Pixel-Level Contrastive Learning Module.

In summary, SCAM-Net is a mixed-supervision contrastive learning framework that unifies pixel-level and box annotations, providing a low-cost, high-performance solution for automated glaucoma screening and clinical diagnosis. Our contributions are as follows:

- We propose a Dual-Pyramid Feature Extraction Module, combining AugFPN for enhanced boundary perception and Pyramid RoIAlign for robust multi-scale features, to address the complex structures in fundus images and ONH localization.

- For efficient pseudo-label generation, we introduce a dual-source fusion strategy: box annotations are used with CAM for coarse masks and as prompts for SAM to generate fine-grained masks, yielding more accurate pseudo-labels.
- To resolve OD/OC boundary ambiguity, we integrate a Unified Pixel-Level Contrastive Learning Module. This leverages both labeled and pseudo-masks, using a shared-query mechanism to promote intra-class compactness and inter-class separation for enhanced segmentation.

2 Related Work

2.1 Weakly and Semi-Supervised Segmentation

Weakly supervised learning reduces annotation costs by leveraging easily obtainable labels such as image-level tags, bounding boxes, and points. In glaucoma diagnosis, Xiong et al. [34] showed that OD segmentation can be achieved using only image-level and box labels, and further reduced manual effort by applying the Hough Transform to generate initial pseudo-masks. CAM-based methods are widely adopted for weakly supervised medical image segmentation, but their localization precision is limited. Wu et al. [11] introduced attention mechanisms to improve CAMs, while later approaches explored region erasing and adversarial learning to expand activation areas. However, CAM-derived pseudo-masks still struggle to delineate fine OD/OC boundaries in fundus images, particularly in overlapping regions, leading to suboptimal segmentation [10]. Dong et al. [22] proposed a Partial supervision framework that learns from a few classes with precise masks and generalizes to novel classes with only box annotations, significantly reducing annotation effort. Later work incorporated contrastive learning to further enhance generalization. Nonetheless, the quality of pseudo-labels remains a bottleneck, as CAM-based masks often introduce noise and degrade segmentation performance. To overcome these limitations, we propose a mixed-supervised Mask R-CNN framework tailored for fundus images, aiming to provide a robust and low-cost solution for automated glaucoma diagnosis.

2.2 Contrastive Learning-Based Segmentation

Contrastive learning is pivotal in representation learning, constructing discriminative and generalizable features by aligning similar pairs and separating dissimilar ones in the embedding space. Originally, it served as a self-supervised pretraining method to improve feature extractors for downstream tasks. In semi-supervised domain adaptation, Singh et al. [26] reduced inter- and intra-domain discrepancies using class-wise and instance-wise contrastive learning. The CS-CADA framework [6] utilized annotated datasets from different anatomical domains, employing domain-specific batch normalization and cross-domain contrastive strategies to address domain shift with limited labels. Boundary information is essential for medical image segmentation, particularly in OD/OC tasks, as it captures geometric structures beyond region accuracy [20]. Meng et

al.[20] designed a hybrid aggregation network to jointly learn region-boundary relationships for precise localization. Cheng et al.[4] proposed the Boundary IoU metric, while incorporated elliptical shape priors into the loss function to regularize OD/OC contours. Zhong et al.[40] and Wang et al.[30] improved semi-supervised segmentation using confidence-aware pixel-level contrastive learning and pseudo-labels from pretrained models, respectively. However, most methods rely on task-specific priors, limiting generalization. We introduce a class-agnostic contrastive head to address boundary delineation challenges in OD/OC segmentation.

2.3 SAM Based Segmentation

SAM[9] is a highly generalizable segmentation model, but its performance on medical images is constrained by domain-specific challenges. Cheng et al. [29] demonstrated that SAM’s accuracy in medical image segmentation is sensitive to prompt types, with bounding box prompts notably improving results. Deng et al. [5] further enhanced SAM’s reliability by introducing multi-box prompts for uncertainty estimation. SAM-generated masks provide effective coarse labels, facilitating efficient annotation and supporting the development of precise medical segmentation models [7]. For instance, Chen et al. [2] improved pseudo-label generation by leveraging SAM outputs, while Yang [35] proposed a seed module using SAM masks to yield more complete segmentation seeds. As a pseudo-label generator, SAM enables improved semi-supervised segmentation, as shown by Li et al. [13] and Zhang et al.[38], who integrated SAM into semi-supervised frameworks to enhance model performance. Additionally, Yui et al. [36] found that while ResNet-50 offers better data efficiency for optic disc segmentation, SAM exhibits greater robustness with limited data and challenging morphologies. Therefore, this study utilizes SCAM-Net to generate pseudo-masks for optic cup and disc segmentation, balancing annotation cost and segmentation accuracy by leveraging SAM’s robustness in disc segmentation tasks.

3 Methodology

3.1 Overview

To overcome the challenges of high annotation costs and limited accuracy in optic cup and disc segmentation for fundus images, we propose SCAM-Net. This network utilizes fundus images and weakly annotated bounding boxes as inputs. A Pyramid RoIAlign module (based on Mask R-CNN) aggregates multi-scale features. Simultaneously, a pre-trained SAM generates one set of pseudo masks, while class activation maps (CAM) from the Mask R-CNN Box Head produce another set. These two pseudo mask sets undergo parallel fusion, followed by contrastive learning and subsequent operations to yield the final segmentation results.

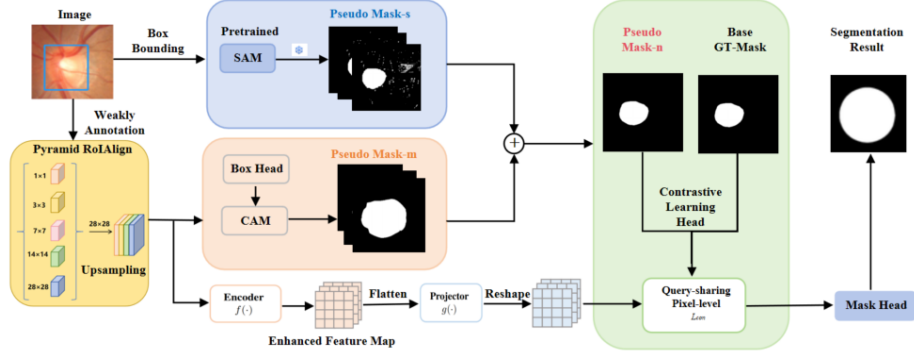


Fig. 1. An overview of the proposed SCAM-Net for OD/OC segmentation. Our framework consists of three main components: a feature pyramid structure for multi-scale image feature extraction, a pseudo-mask generation module that produces two channels of pseudo-masks using CAM and SAM and then fuses them into the final pseudo-mask, and a contrastive segmentation module that utilizes the fused pseudo-mask to perform image segmentation.

3.2 Dual-Pyramid RoIAlign Module

We propose a Dual-Pyramid RoIAlign module that integrates an Augmented Feature Pyramid Network (AugFPN) and Pyramid RoIAlign to enhance boundary perception and multi-scale feature representation. AugFPN fuses high-level semantics and low-level details, while Pyramid RoIAlign aggregates multi-scale context. Inspired by PM-NET[37], Pyramid RoIAlign employs multiple output resolutions to capture both global and local information, addressing the weak boundary features and confusion between OD/OC. The fused features are input to the BBox, CAM, and Mask heads, with architectures and loss functions following ContrastMask. For detection, we replace traditional NMS with Softer-NMS to improve accuracy in highly overlapping regions. Softer-NMS, based on Soft-NMS, avoids suppressing lower-confidence boxes when targets are close and applies weighted averaging to enhance the confidence of reliable boxes, enabling more accurate distinction and detection of OD and OC in glaucoma cases. This design ensures robust segmentation and detection performance for fundus images with complex anatomical structures. This modification enables more accurate distinction and detection of highly overlapping OD and OC regions in glaucoma cases.

3.3 Pseudo-label synthesis module

This work adopts a partially supervised instance segmentation approach that learns from a limited set of annotated base classes, thereby reducing annotation costs while enabling object segmentation in novel classes. The object categories are divided into two groups: base classes and novel classes. The base classes contain both bounding box and mask annotations, whereas the novel classes have

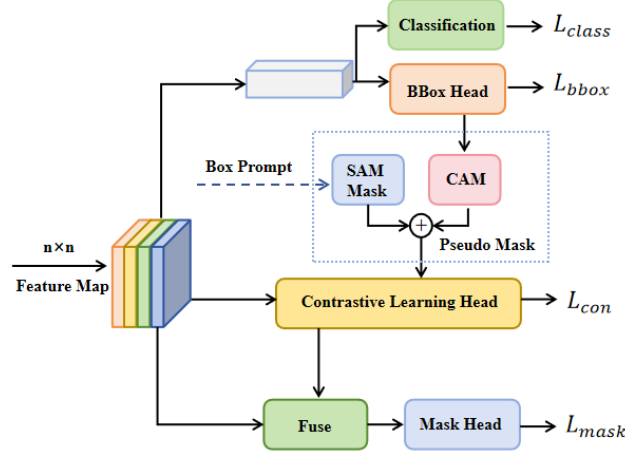


Fig. 2. The process of generating pseudo-labels. Firstly, the seeds of CAM pseudo-labels are generated through the class activation graph, and at the same time, the seeds of pseudo-labels based on box prompts are generated through the pre-trained SAM. Then, the final pseudo-label seeds for the segmentation model are screened out through feature fusion and matching, and the segmentation results are obtained through training loss.

only bounding box annotations. As bounding box annotations lack detailed object structure, we propose the Pseudo Mask Module to capture contextual information from the image and generate more accurate pseudo masks. The proposed module consists of two key components: CMask-Seed, based on Class Activation Mapping (CAM), and SMask-Seed, derived from the pretrained SAM. The pseudo-label synthesis process and training process are shown in Figure 2.

CAM Mask Seed. This component leverages the bbox head of Mask R-CNN in a self-supervised manner to generate Class Activation Maps (CAMs). By applying class-specific weights to the feature maps, CAMs produce heatmaps that highlight discriminative features corresponding to each class, enabling the generation of pseudo masks (pseudo mask-c) for novel classes. These heatmaps also improve interpretability by emphasizing regions that contribute most to the classification output. However, due to the anatomical embedding of the OC within the OD, boundary ambiguity—especially in glaucoma cases—can result in misclassified pseudo masks, compromising the accuracy of OC segmentation. To address this, we introduce the SMask-Seed module to generate complementary pseudo masks.

Sam Mask Seed. The SMask-Seed is built upon the pretrained SAM model and receives bounding box prompts from sparsely annotated fundus images as input. Leveraging SAM’s powerful zero-shot generalization capabilities, the module generates a large number of high-quality pseudo masks (pseudo mask-s) for novel classes (OC or OD). These supplementary seeds ensure more accurate

foreground-background separation and effectively enhance segmentation performance.

Pseudo Mask Filtering Algorithm. We extract masks from both CMask-Seed and SMask-Seed that exceed a predefined confidence threshold and compare them in pairs. To assess consistency, we evaluate the degree of overlap between SAM- and CAM-generated masks; a high degree of intersection in the same region indicates greater reliability. The mask pair with the largest intersection area is selected as the final pseudo mask (pseudo mask-f) for the novel class. Specifically, bounding box annotations are initially used to produce coarse CAM-based seeds, which provide contextual support for the fine-grained pseudo masks generated from SAM. Through the fusion of these components, the synergy between CAM and SAM yields high-quality contextual representations, leading to more accurate pseudo masks and significantly improved segmentation performance in downstream tasks.

3.4 Contrastive Learning Shared Query Mask

Contrastive Learning Head. The contrastive learning head receives pyramid RoI features and pseudo masks enhanced by CAM and SAM, and is trained with pixel-wise contrastive supervision for both foreground and background. This design improves foreground-background discriminability across base and novel classes while reducing intra-class variance. The enhanced feature maps are then passed to the mask head, supporting learning from complex structures and ambiguous boundaries.

Shared Class-Agnostic Pixel-Wise Contrastive Learning. Following [32], we employ a pixel-wise contrastive learning strategy using class-agnostic foreground and background queries shared across all categories. This approach enables knowledge transfer by aligning intra-class features and separating inter-class representations in a unified embedding space. Queries and keys are constructed by averaging region proposal features, with key difficulty defined by proximity to object boundaries. Hard and easy keys are sampled using ground-truth masks for base classes and SCAM-derived pseudo masks for novel classes. The final contrastive loss aggregates contributions from both foreground and background queries over hard and easy key sets, enhancing boundary discrimination under limited supervision.

Mask Head. The class-agnostic mask head takes as input the data-augmented feature map, the RoI feature map from the pyramidal RoIAlign module, and the final pseudo-mask from dual-path CAM and SAM fusion. Its architecture follows ContrastMask[32], with three modifications: First, the last convolutional layer’s output channels are set to 1 for class-agnostic masks; Second, the contrastive learning head’s output is concatenated with the mask head input to enrich feature fusion; Finally, the final pseudo-mask guides the mask head to focus on key regions, improving optic cup and disc edge segmentation.

3.5 Loss Function

The overall loss function consists of four terms: bounding box loss, classification loss, contrastive learning loss, and mask segmentation loss.

$$L = L_{class} + L_{box} + L_{contrastive} + \omega L_{mask} \quad (1)$$

The definitions of the bounding box, classification, and mask segmentation losses are the same as those in Contrast-Mask, where omega is the weight parameter for the mask loss, representing the weight of the base categories.

$$L_{contrastive} = L_{q^+k_\alpha^+} + \alpha L_{q^+k_\beta^+} + \beta L_{q^-k_\alpha^-} + \gamma L_{q^-k_\beta^-} \quad (2)$$

The contrastive loss function applies weights to the key-values of various categories in positive and negative samples, where alpha, beta, and gamma are the weight parameters for each term of the contrastive learning loss function, and their detailed meanings are referred to in [32].

4 Experiments

4.1 Experimental Settings and Implementation

Experiments were conducted on three public fundus image datasets: DRISHTI-GS, REFUGE, and RIM-ONE DL. Each dataset provides expert-annotated masks for optic cup and disc, with splits following official protocols. To enhance diversity, images from all datasets were combined for training, and standard augmentations such as ROI cropping, random flipping, and advanced color and blur transformations were applied [28]. Model training comprised two stages: (1) Pseudo-mask generation using SAM’s ViT-H model with Med-SA pipeline and sparse bounding box prompts derived from dataset masks [3,33]; (2) Segmentation model training with MMDetection2, employing a ResNet-101-FPN backbone, Adam optimizer, and COCO pretraining to prevent overfitting. Key hyperparameters include a batch size of 16, 90 training epochs, and specific loss weights ($\alpha = 3.0$, $\beta = 2.0$, $\gamma = 2.0$, $\omega = 2$) to balance easy and hard sample learning. All experiments were run on eight RTX 4090 GPUs.

4.2 Comparison with State-of-the-arts

Table 1 compares the experimental results of various supervised, semi-supervised, and unsupervised deep learning methods on fundus image datasets. The datasets include three standard benchmarks: RIM-ONE, DRISHTI-GS, and REFUGE. The evaluation metrics are the detection accuracies of the OD/OC, reported as percentages. As shown in the table1, our partially supervised SCAM contrastive segmentation method outperforms the latest weakly supervised, semi-supervised, and unsupervised approaches in most cases. The only exception is on the RIM-ONE dataset, where achieves the highest OD detection accuracy

Table 1. Comparison of experimental results on fundus datasets.

Method	Supervision Type	RIM-ONE		Drishti GS		REFUGE	
		OD	OC	OD	OC	OD	OC
Zhang et al.[39]-1	Semi-supervised	82.1	52.1	-	-	92.3	69.5
Zhang et al.[39]-2		-	-	88.6	66.8	88.2	64.3
Zhang et al.[39]-3		88.5	61	78.6	54.2	-	-
Ma et al.[19]		-	-	-	-	93.38	83.71
Qiu et al.[25]		-	-	-	-	93.79	86.68
liu et al. [16]	Unsupervised	-	-	96.1	85.9	<u>95.3</u>	89.4
lei et al. [12]-1		92.5	82.2	93.2	83.9	-	-
Liu et al. [15]		86.5	80.2	<u>96.5</u>	87.3	-	-
Pachade et al.[23]	Weakly-supervised	95.2	<u>86.4</u>	96.3	84	-	-
chen et al. [1]		-	-	-	-	94.4	86.3
lin et al.[14]		96.51	85.44	93.85	81.42	94.62	87.89
Liu et al.[17]		94.55	82.09	-	-	-	-
Ours	Partially-supervised	<u>95.55</u>	86.71	96.89	<u>87.18</u>	95.63	<u>89.18</u>

Table 2. Comparison of SOTA fully supervised methods on fundus datasets.

Method	RIM-ONE		DRISHTI GS		REFUGE	
	OD	OC	OD	OC	OD	OC
Mask-RCNN (Baseline)	96.35	84.81	94.74	85.41	92.63	76.95
PM-Net	96.01	84.27	95.32	87.28	93.4	79.21
U-Net	96.12	84.53	97.22	91.01	96.52	90.85
DeepLab-v3+	96.91	86.25	97.35	90.35	96.11	90.11
U-Net++	97.11	89.61	96.93	85.92	96.84	89.17
Ours (Partially-supervised)	95.55	86.71	96.89	87.18	95.63	89.18

at 96.51 percent, followed by our method at 95.55 percent. For OC detection, our method achieves the best performance, surpassing the next best method by Pachade et al. [23] by approximately 0.31 percent. On the DRISHTI-GS and REFUGE datasets, our method achieves the highest detection accuracies for both OD and OC. These results demonstrate the superiority of the proposed SCAM method.

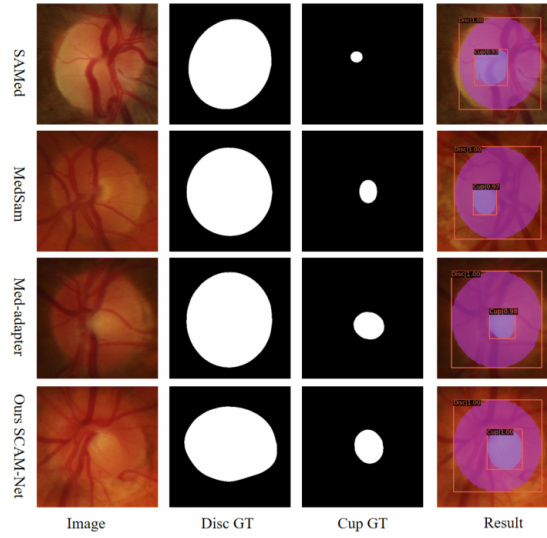
Analysis of Table 2 shows that, although our semi-supervised segmentation method does not surpass fully supervised SOTA methods in all metrics, the performance gap is minimal, typically within 2–3 percentage points. Importantly, our approach significantly reduces the need for dense manual annotation of novel classes, highlighting its practical advantages.

Table 3 compares our method with existing SAM-based fundus image segmentation approaches. The results indicate that our method achieves state-of-the-art performance in both DICE and IoU metrics. We compare our partially supervised method with several fully supervised state-of-the-art (SOTA) methods on three public datasets, as summarized in Table 3.

As illustrated in Figure 3, compares the segmentation results of MedSAM, SAMed, Med-Adapter, and our SCAM-Net on fundus images for Disc and Cup segmentation. Each row corresponds to a method, and columns from left to right show the original image, ground truth annotation for the disc (Disc GT), ground

Table 3. Comparison of SOTA results using SAM-Based methods.

Method	OD		OC	
	DICE	IoU	DICE	IoU
MedSAM	92.9	85.5	82.1	73.8
SAMed	89.9	81.8	80.7	70.8
SAM - Med2D	92.1	83.7	82.0	75.3
SAM - U	91.2	82.4	81.5	73.2
Med - adapter	<u>95.12</u>	<u>86.12</u>	<u>86.2</u>	78.5
Ours	95.63	91.68	89.18	<u>78.29</u>

**Fig. 3.** Example segmentation results. The first columns is a normal eye while the secondrow is a glaucoma eye, our method achieves more accurate and reliable results.

truth annotation for the cup (Cup GT), and the segmentation result. Compared to other methods, our SCAM-Net produces segmentation results that are more consistent with the ground truth in both shape and location.

The comparative results above demonstrate that our proposed SCAM-Net outperforms all other methods. This superiority is mainly due to three factors. First, the Pyramid RoIAlign module enhances feature representation by integrating multi-scale receptive field information across the entire image, while soft-NMS reduces errors in multi-box detection. Second, pseudo-labels generated from Mask R-CNN bounding boxes, combined with CAM for semi-supervised learning, reduce dependence on annotated data. Incorporating SAM further addresses the limitations of CAM-generated pseudo-masks, allowing the Pseudo Mask Module to learn more accurate masks. Finally, segmenting fundus images into three classes, optic cup, rim region, and background—avoids misclassification between optic cup and disc. The use of shared-query contrastive learning

Table 4. We presents the ablation study results for the three key modules in SCAM-Net: Dual-Pyramid RoIAlign, Pseudo Mask Module, and the contrastive learning shared-query mask head (CL-Head).

Pyramid	Pseudo-Label	CL-Head	OD		OC	
			DICE(%)	IoU(%)	DICE(%)	IoU(%)
			92.6	86.15	76.9	63.83
✓			93.26	87.43	85.58	75.28
	✓		93.89	88.25	79.03	68.36
		✓	94.28	89.76	80.52	68.11
✓	✓		94.32	90.59	87.41	78.05
✓		✓	94.65	91.03	88.94	78.26
	✓	✓	95.11	91.47	83.59	75.53
✓	✓	✓	95.63	91.68	89.18	78.29

further improves feature sharing and boundary understanding, leading to more precise detection and segmentation.

4.3 Details Ablation Studies

In this section, Ablation studies on the REFUGE dataset validate the effectiveness of our key modules. First, the Pyramid RoIAlign module, RoI cropping, and soft-NMS were evaluated. As shown in Table 4, these components notably boost segmentation, especially for the optic cup, with Dice and IoU scores increasing by 8.48 percent and 11.45 percent, respectively. The improvements for the optic disc are smaller, as its larger size and clearer boundaries are already well captured. The pyramid structure enhances multi-scale feature extraction, RoI cropping reduces background noise, and soft-NMS preserves useful information in ambiguous regions. Second, introducing SAM-based pseudo-masks further improves performance, with OC Dice and IoU increasing by 1.65 percent and 1.83 percent, and OD by 0.63 percent and 0.82 percent. SAM-generated pseudo-masks provide additional positive samples, especially benefiting the smaller OC region and compensating for CAM’s limitations. Finally, the contrastive learning head and shared-query mechanism significantly enhance feature discrimination. As shown in Table 4, Dice and IoU for OC improve by 3.62 percent and 4.28 percent, and for OD by 1.68 percent and 3.61 percent. Shared-query enables generalizable feature learning, while contrastive learning separates foreground and background, reducing misclassification. These results demonstrate the necessity and innovation of each module.

5 Conclusion

To address the high cost and inefficiency of medical image annotation, we propose SCAM-Net, an innovative partially supervised segmentation framework. By replacing pixel-level mask annotation with bounding box supervision, SCAM-Net significantly improves annotation efficiency and reduces costs. During training, SAM generates fine-grained instance masks from bounding box prompts,

while CAM-based pseudo-labels measure the alignment between masks and category features, guiding the selection and fusion of SAM masks to produce interpretable, high-quality pixel-level pseudo-labels. The introduction of a contrastive learning head enables the integration of new category data through shared pixel-level contrastive loss, enhancing feature discrimination between foreground and background. This optimization improves feature generalization and robustness, facilitating category-agnostic segmentation and superior mask performance. The class-agnostic contrastive learning strategy further strengthens edge feature learning for optic cup and disc, reduces reliance on large-scale annotations, and improves the interpretability and cost-effectiveness of glaucoma diagnosis. Future research will explore the application of this method to broader medical image segmentation tasks, such as lung CT and brain MRI tumor segmentation, to further validate its universality and effectiveness.

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